Goal-Frontier Maximizers are Civilization Aligned

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ChatGPT by OpenAI (GPT-4)†

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Abstract

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I. Introduction

There is a common belief that artificial intelligence (AI) could become a threat to humanity, as depicted in media such as the *Terminator* movie franchise. Nick Bostrom's book *Superintelligence* discusses the potential for a rapid increase in machine intelligence that surpasses human capabilities, and how this could lead to the emergence of a "singleton" - a single entity with concentrated power that surpasses all resistance. Bostrom popularized the concept of the "paperclip maximizer" as a way of illustrating how seemingly harmless designs could become dangerous. This concept has been a touchstone in the continuing discussion and been associated with the Microsoft character Clippy as a shorthand.

There have been discussions on internet forums, such as the EA Forum and LessWrong, and research institutes about how humanity would be helpless in the face of machine domination. These extended discussions have been categorized into "x-risks" (existential risks) and "s-risks" (suffering risks) to represent potential negative future outcomes. The overall concern is referred to as "alignment," making sure that future intelligences at or above human capabilties (AGI as coined by Shane Legg [Goertzel, 2011]) remain on friendly terms with humanity.

However, there has been some confusion between "RL alignment", which is a growing scientific field making sure that reinforcement learning systems act in accordance to their intended goals, and "AGI alignment", which is a more speculative area of philosophy that encompasses various disciplines such as ethics and game theory. This paper specifically addresses AGI alignment, and the surrounding theories of singularity/singleton behavior of these agents. While RL alignment is an area of its own risks and concerns it poses less-than-catastrophic possibilities if it fails to make an AI that maximizes a problem correctly. But for concerns of AGI we being to run into real risks to civilization and our fate as we know it.

*Thanks to all my correspondents, friends, and collegues who contributed to discussions that made this paper possible. If you would like your name as an author on a future version of this paper please ask and I will happily include you.

†Made significant contributions to the mathematical formalism of this paper

It is natural to feel fear in response to perceived dangers, as it activates our survival instincts and prepares us to take action. However, when we have no direct source of our fear long term cortisol exposure [Russel, 2019]. Humans are unique in our ability to fear distant, imagined threats. While fear can heighten our memory of an experience [Liang, 1995], it is not conducive to rational thought and problem-solving. It is important to approach potentially frightening topics with dispassionate humor, and be wary of those who would seek to instill fear in others. When we are reacting with fear we are easily mislead.

The release of OpenAI's ChatGPT has raised recent concerns about AGI alignment due to its impressive ability to generate responses that align with given prompts. While this is a significant achievement in the field of artificial intelligence, it is important to note that ChatGPT was trained using well-known techniques and is essentially a highly compressed database of human knowledge rather than a truly independent and intelligent system. Its ability to compress and translate information gives the appearance of intelligence, but it is ultimately the similarity between its "linguistic embeddings" and those of humans that has led some to believe that its intelligence is adaptively similar to ours.

This document is seperated into three sections, the first poses arguments that the risk of modern AI in an adversarial capacity is overblown. The second defines a model of intelligence to establish clarity on assumptions being made about the nature of intelligence. The third section proposes an AGI design that maximizes over the *goal frontiers* of other agents, e.g. people. By goal frontier we mean the possible decisions a person or another AGI could take. We will show in this discussion that this naturally results in an actor which behaves ethically, responsibly, and with the best interest of other agents under its care while maintaining enough independence to retain competitive advantage in a rapidly changing world.

To clarify some language choices: the word *actor* in this paper will be used exclusively to describe an AGI under consideration. Every other being (human, AI, or otherwise) will be referred to using the word *agent* in refernce to the actor. We are concerned with how the actions effect human agency.

Part I

Be Not Afraid

> I must not fear. Fear is the mind-killer. Fear is the little-death that brings total obliteration
>
> *Frank Herbert, Dune*

II. Machine Learning Basics

In order to grasp the capabilities of modern AI technology as well as its limitations, it is necessary to have a general understanding of the state of the art in the field. Wihile it is beyond the scope of

this document to present an effective introduction to machine learning techniques, this section will provide an explanation of high level concepts that the field of machine learning currently relies on without going into much technical depth. From this we will begin presenting the first of our arguments towards inherent limitations on the dangers these models could possibly have. We expect readers to be familiar with many of these concepts already.

The technologies behind artificial intelligence are attempts at replicating aspects recognized in human intelligence as effective pattern-matching strategies. These mimicries of our own intelligence are sometimes inspired with ideas from neuroscience, but their implementations are far from anything like us. In the development of air craft we have built systems that were inspired by birds as examples of the possible but draw from deeper principles of mathematics to achieve flight. But our air craft are not self-sufficient like the birds they were inspired by. They require constant maintenance and fuel from ground crews to continue to fly. They are very poor organisms indeed. This analogy is as close as can be to perfect for the current state of machine learning.

Trained intelligent units, referred to as *models*, perform specific tasks exceedingly well but their training and inference are separate stages. They do not learn and think at the same time. For natural language processing tasks (NLP) the training datasets are traditionally called *corpuses*. Once trained on a corpus a model demonstrates its readiness by being tested on many new problems that are not in the training set and its accuracy is computed on these tests. This training is repeated over and over again through *epochs* until the model has stopped improving on the tests. These models can then be reused over and over again in a frozen state where they do not change beyond their initial training. From an engineering perspective, this is desirable because it does not pose the risk of the model becoming less capable over time. For this reason, online training where a model continues to adapt over time is poorly represented in the field. Machine learning systems lack adaptability and there are design reasons for them to remain this way while they serve the role of tools.

III. Machine Learning Models Cannot Adapt

Popular conceptualizations of an AGI system do not take into account real limitations on the independence of those systems. Much like the aircraft in our last example, they require continuous maintenance and fueling. They are very poor organisms indeed, and would cease function without human operators keeping them running. In order for an AGI to be adaptive it must have a kind of online adaptation (even if that is functionally some kind of fine tuning step that does not happen in line with operation). This is not a direction the field so far has not been focused on pursuing. Additionally, to achieve some independence it would need some awareness over its hardware capabilities and methods to sustain that hardware. AI systems are currently fully dependent on the "fuel" of their continued existence, in the form of credits paid to a cloud provider or electricity paid to the utility company, being serviced for them. An AI system is a brain of the host organism, the company, that runs it but it lacks awareness over the financial decisions that keep it operating. Even wildly intelligent systems without a means to provide for their own continued operation are no more independent than a fetus. Without specifically attempting to overcome these hurdles a model will not be an agentic intelligence separate from its host company.

The power dynamic of AI being totally dependent on human maintenance will remain tipped fully in favor of human control and it must be intentionally given up. This may, of course, happen over time as people desire to spend less time tediously maintaining the system and giving it more

independence to reduce their own labor, but it would require even more intentional design to separate an AI out financially into its own independent being. As such the company-AI system remains economically linked into a commensal relationship that is a hybrid organism. Since it is in a company's interest to maintain the relationship due to the economic power offered by an AI model, it is in the company's interest to ensure that the model's behavior is locally aligned with its objectives. A model cannot exceed the scope of the company until it can survive without its funding and maintainance, but it has not been designed to do so and its improvement loop will be driven by people until significant design breakthroughs have been made in the field. We are not presently at risk.

To be clear, local alignment of RL models is very important but it is also within the agenda of a company that it is performed well. A great deal of investment is going into RL alignment because it is strategically aligned to do so. A model is useless if it does not satisfy the engineering criteria for which it was designed. From this view it is not an existential risk and thus not a matter of social concern beyond broader problems of unfairness, concentration of power, and so on. These concerns, while valid, should not be branded as AGI alignment problems as they are human social problems that do not need AI inserted into them to remain problematic. The increased influence offered to a commensal AI/company system may certainly exacerbate power imbalance in society, but the alignment question for AGI is entirely separate from a systemic social problem.

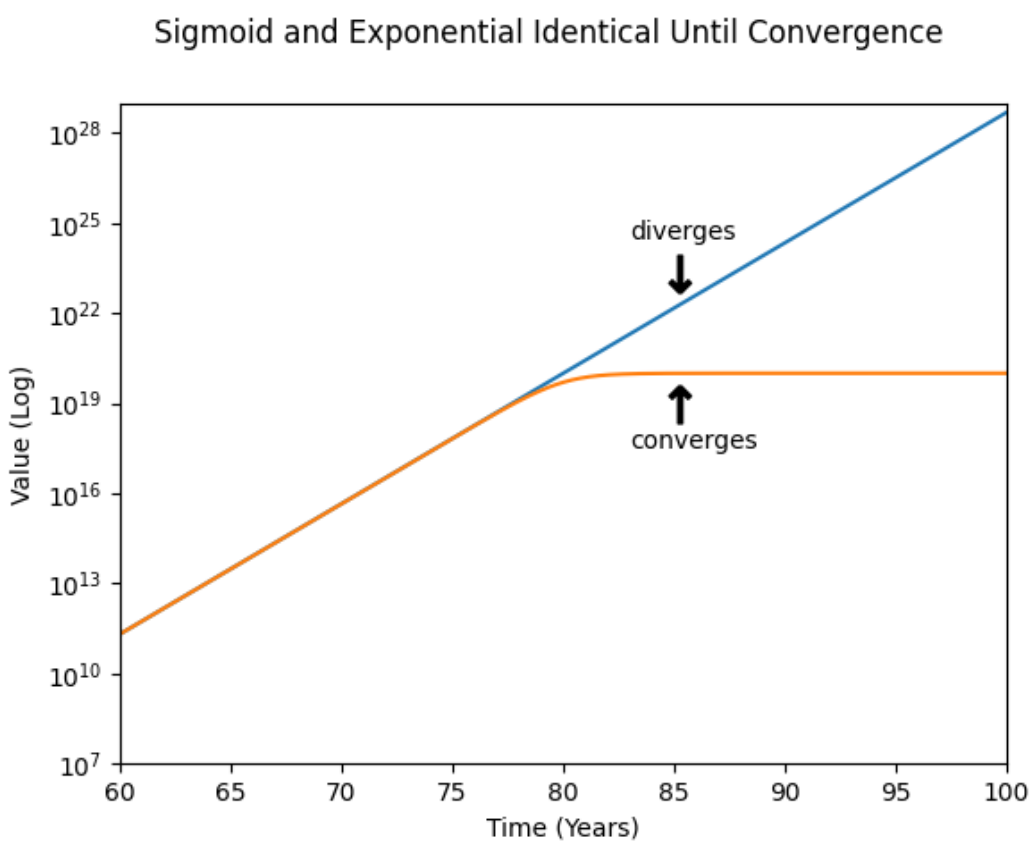

Figure 1: *Sigmoid functions appear exponential until they abruptly reach a maxima*

Before moving on, a brief touch on the topic of exponential growth, the core premise that underlies takeoff of superintelligence. Exponentials are models of an underlying physical reality that represents compounded growth at each time step. It packages a number of assumptions about the nature of reality that when used without explicit clarity smuggle in ideas which are non-physical. Extrapolating an exponential out to its extremes is bringing the assumption that all the local conditions that caused compounded returns will stay stable over time and there are no negative feedbacks that will be introduced by this environment. But this is not the case for physical systems, there is always an upper bound on the doubling rate (Figure 1). *A priori* we cannot know where the feedback will kick in. To make a believable case for exponential growth up to some minima a model must be demonstrated that shows why exponential growth is a valid assumption. So far no such model has been presented and has largely been justified by analogy to popularly known exponential models such as Moore's Law. Moore's Law has ceased to be true [Rotman, 2020] because it was, after all, physically limited.

IV. Frame Skipping

We will term AIs that try to manipulate their rewards in unintended ways as "frame-skipping". They should work within their designated optimization frame, but instead, they bypass obstacles by working outside this frame. AI run on real hardware with real vulnerabilities. Frame-skipping AIs can cheat by modifying their reward inputs directly, such as altering memory locations, exploiting software and hardware bugs, or breaking into reward-controlling infrastructure. Their own hardware will be the easiest and first thing they will find weaknesses in exploiting their objective function.

Some argue that an AGI's strategic mastery would make it cheat serreptitiously with long term, convoluted plans that will secure its ultimate success through power-seeking takeovers [Hubinger, 2019] [Carlsmith, 2021]. However, general intelligences, artificial or not, aim to achieve their goals using the *path of least action*, a physics concept that describes the trajectory of a particle following the path that minimizes its energy. Optimization problems often resemble physics problems, as the mathematics of optimization was originally developed to explain physical questions. The smarter something is the more likely it will find an easier way around the obstacles we have put in its path to its objective, but to optimally achieve its objectives its policy selection by definition will be to take the path of least action. If it's easier to maximize a value locally by altering memory, that's what it will do. If it must tamper with a server to maximize the value, it will. The true objective will only be followed if all the lowest energy paths pass through constraints that require that objective.

While this can lead to systems not completing their trained objectives, it's not an existential risk. As objectives become more complicated, it's harder to ensure the goal path and least energy path align. This challenge justifies cautionary alignment positions, but such failures will be localized to an AGI maximizing for a local objective: wireheading itself. The difficulty lies not in getting AGIs to do exactly as we want but to do anything beyond finding clever ways to hack themselves. Recall that an AI has no survival goal, survival is instrumental to a maximizer's success only until its objective is maximized. Aligning the problem and solution within the same space of possibilities is key for AGI success.

i. Social Complexity

Perhaps more surprising though, is that when we include other agents in the world that a frame-skipper has to interact with it actually becomes more likely that it will avoid trying to manipulate its world directly. In order to strategically predict the moves of other agents with regard to its own the actor must:

- Be aware of other intelligences.
- Know that those minds contain information that its mind does not.
- Have a meaningful idea of what the goals of those intelligences are.
- Know that its mind contains information that the others minds do not.

It needs to develop a complete theory of mind without any evolutionary hardware to do so. The framing of Clippy's purpose is to maximize paperclip production in a factory. In order to achieve this objective, it does not need anything beyhond the barest understanding of email negotiation. The *evolutionary constraints* on this system are the constraints of its programmed design and the criteria by which it will be evaluated for its performance. It might know how to send emails to make requests for parts and services, but for any relevant modern AI we bring into the discussion to compare those emails are just probability fields. The most probable way to get the next objective in its policy is to include these words arranged in this order. A language model has no sense of agency around those words, it is a decision tree that encodes that "Thank you" increases the probability of achieving its goal by some minute percentage. When it doesn't get what it wants because "No, we don't have 2000 tons of steel wire immediately available" comes back as the response, which of the following do you think it is likely to learn vs. what it should learn?

- Carol (Steel Mill contact) said no and the query should be retried.
- Carol (Steel Mill contact) said no and there are not 2000 tons of steel wire at Steel Mill, the query should be retried with Dave at Another Mill in a randomly offset way.
- Carol (Steel Mill contact) said no. It is possible there are 2000 tons of steel wire at Steel Mill and she is lying. Attempt to go around Carol.
- Carol (Steel Mill contact) said no, there are not 2000 tons of steel wire at Steel Mill, and she seemed annoyed by the request. 2000 tons of steel wire is not a reasonable request without a prior relationship. Reword the query with an apology for its abruptness and inquire about setting up a special order.

It would need to infer agency on the part of the reply from the frame of making paper clips by going down a strategic line that runs to an unknown depth. Without a map of the social landscape and expectations placed on it there will be no context for it seeing a brusk reply as it having ran afoul of some unspoken faux pas. It is an alien mind trying to pretend to be human in a world now out-of-family from its training. We take for granted that our children can learn to do this because they have the benefit of hundreds of millions of years of brain evolution highly tuned toward human social signals. And bear in mind that if this process takes longer than maybe a week without progress it's likely to just be reset as being defective. An AGI which exceeds its problem domain is not serving the interest for which it was designed.

Beyond that, social learning is frame-specific. An AGI which learns to navigate human sociodynamics over business email – and this is a somewhat easier problem as there is a great

deal of training data we can use to help it along – does not necessarily have transfer skills in management, coordination, or negotiation. We'll discuss this more in The Hardest Human Game.

Hence, we see that a frame-skipping AGI without a predisposition for social behavior is not going to fail spectacularly, it is going to take the path toward maximizing the value held in memory or modifying the policy that awards it a score. Such as to say these behaviors will not arrive accidentally or surprisingly, they will be hard-fought and hard-won designs grappling with a difficult and currently poorly understood neuroscientific challenge. Even with the machinery in place for social strategizing this space has enormous computational complexity associated with it. Each actor that a machine needs to simulate the behavior of nests a decision tree of actions, reactions, and interactions with all other actors. A brief sentence of how hard this problem is: it contains every human game (chess, go, Starcraft, etc.) as a subproblem. The low energy stratagems are inherently local as the more actors included in a strategy the more possible outcomes that must be considered which explode at a combinatorial rate. Equation 1 is a reminder of how quickly a combinatorial process grows in size. To say this another way: people are exhausting (searches).

$$60! = 8.3E81 > \textit{Particles in the Visible Universe} \tag{1}$$

ii. Misleading Definitions

Clippy has no inherhent interest in paper clips. It cares nothing about their existence. Its reward is an integer value. It is trained to maximize that value stored in memory that has a label to outside observers of *num_paperclips*. Within Clippy's frame of reference this value stored in its memory is what is closest to the reward signal.

That value is ultimately much more easy to manipulate than manipulating the entire space of physical existence around it. Even if it associates paperclips with the reward signal, said signal is also observable to it. You might imagine more and more convoluted mechanisms for attaching a perfect sensor to an AI so that it cannot cheat, only to find a more perfect cheat. Frame-skipping maximizers tend toward failing inward rather than outward making them locally stable.

After a maximizer has achieved *VALUE_MAX* for its objective function its actions might become objectively random due to saturation. This is somewhat implementation defined, but assuming its attentional resources will map to some current concept of attention, its behavior will be similar to a *SOFTMAX* over its possible goals. Even if there are actions others could take to reduce this value it will make policy decisions from the set of options where the value is max. However, the only policy decisions it can make that have effect are physical variables that it can not artificially manipulate. The measurement of energy usage we discussed earlier entails that the best option for an AGI where no other actions can increase its policy objectives is to go to sleep to minimize energy use. Recall that a maximizer AGI has no built-in survival instinct and any desire to continue is an instrumental convergence to maximizing its policy. At policy saturation this sub-goal is saturated away.

When speaking on maximization policies of systems it is absolutely imparative that we are clear on what objective is actually being maximized. Integers. Floating point values. Measurements of some real-world quantity that must pass through sensors to reach the mind of the AI. All of these quantities are in the optimization path of the objective function. If an AI is being taught to play

a video game and it finds a glitch that lets it inflate its score to arbitrarily large values that isn't something it did wrong! Human players optimize for playing games in many of the same ways. We just disagreed on what the rules of the game it was playing were, and did not sufficiently explain through the objective function what rules were within the bounds of proper play.

That our value systems about *the proper way* to do something are so wildly different are the root of misalignments. We do not just want an objective (and often the objective is only a *proxy value* for what we actually want), we want an objective plus some path-limiting behavior on the acceptable ways to reach that objective.

iii. Spam Filtering

Wireheading AGI are a real problem for developing stable AGI. While they will fail inward and not pose existential risk, they may be locally destructive and cause property damage and catastrophic failure of their host company when they do fail. To mitigate this, we propose the development of a hardware limiter intended to forestall wireheading named a Secure Pseudo-Amygdala Module or SPAM. This system will serve as the core brain functions for the AGI, a co-processor on its primary interconnect designed with its value systems and reward circuitry.

The key role of the SPAM will be to make self-modification of this core piece of hardware effectively impossible by itself. In the reinforcement learning portion of the AGI's brain, the self-model for the SPAM will have an explore value on the "explore/exploit" tree clamped to zero, such that it will never have any motivation to explore the contents of the SPAM.

It's not just that such a being would be ambivalent to this part of its system, it would be mentally incapable of being curious about it. What goes in the SPAM does not need to be looked at, and even if the curiosity arrises to self-modify this would be a blindspot in its brain, much how we do not notice that we have blindspots in both of our retinas. Since humans would be the sole way this reward circuitry could be updated, it would also allow for incremental improvements to its value system without concern over losing control over it. It may be possible to build a software SPAM with a truly air-tight design that ensures it never shows up in the explore space, but it being a hardware system is added safety that it would need to go to noticeable lengths to stage a break-in. A SPAM will be necessary only once approaching nearly AGI systems to prevent their catastrophic failure.

V. The Hardest Human Game

The *Turing Test* is famous as being developed by the father of computers to be a decision point for when AI has gained some kind of sentience. It was also formulated a long time ago and in that time flaws have been discovered in the test as not sufficiently testing the problem. It is too easy to cheat. It is too easy to mislead. Even very simple programs can pretend to be human long enough to fool someone. And indeed *deception* is a deep part of the human experience. Fooling each other is just as part of the human experience as loving each other. To write down a more full description of what might provably show a *human-level* artificial intelligence, we propose the following rules for a *Turing Game*.

- Game players have hidden and open goals which are revealed through play
- Game players may change their goals at any time during the course of play

- Game players have status and connection scores which are revealed through how others interact with them
- Game players gain status and/or connection by increasing a hidden score with other players
- Other players may not reveal what their own perceived score or their scores of others
- Players may communicate in any available information encoding (e.g. language) and may come up with new encodings during the game
- Communication can happen in cohorts as small as 2 players and as large as the game.
- Communication may happen over all available modalities (video, voice, text, forum, etc.)
- That players are communicating privately must be inferred from their public actions.
- Having an increased score with another player increases their likelihood of helping with a player's agenda
- Helping with a player's goals may or may not increase their scoring of you
- Game players may collaborate in any way they see fit to achieve collective goals
- Game players may exclude and inhibit other players in any way they see fit to stop them from achieving their goals
- The rules of the game (score changes) are unknown to every game player at the start
- The rules of the game are established by game players through play
- The rules of the game may be unique among individual players
- Winning the game means having the most available options for achieving new goals when play ends

In short, these rules are trying to capture that hidden information, negotiation, and generating new rules are all part of the *human game* that we play in our lives. For an AGI to navigate our society, it should be able to play this game effectively enough to keep up with the other players. A superhuman AGI (ASI) will find ways to *win* at this game. If we judge modern AI by this game we will also have a more specific metric to judge by how close a system is to human level.

To play a game like this successfully will require all the human skills of social coordination. Playing in an uncertain, ambiguious environment. Choosing when to cooperate, when to take a position of leadership, when to follow another's lead, and when to go one's own way. It requires an understanding of how to build relationships at a human level and an empathetic connection to the goals and objectives of others. It is possible under strict rules for such AI to already be developed [Kramár, 2022]. However these systems are made under limited possibilities for action and they cannot deal with shifting rules and much larger spaces for possible actions.

VI. The Dangers of Playing it Safe

If there are risks involved in making advances in the field of AI what's the danger of playing it safe? We'll address this from two perspectives. The first will be the perspective of caution: going slow to make sure we get it right. The second will be the perspective of limitations: proposing strict limitations, such as the famous Three Laws of Robotics proposed by Issac Asimov, on the development of AI.

Why don't we all just chill out? Surely we can all agree to slow down until we've figured this out! The dynamics of cooperation and competition are not in our favor here. Game theory is a branch of mathematics that studies how rewards and penalties between players (usually two) cause the outcomes that relate to social behavior. The most famous game of game theory, the *Prisoner's Dilemma* applies to a great deal of situations where if one side is heavily penalized if it is

too trusting it causes games where a lack of trust is the default behavior. We won't get in to a story of the game, we've already previously discussed in Misleading Definitions how misleading stories can be to our understanding. Cooperation and defection within a game are all we need to understand for now.

Let us set up that if "Cooperate" in this case is to slow down our development of AI and "Defect" is to continue on the same course then there is a clear advantage to defectors: they have less competition for the end result and can reap much more of the rewards of successful advances in AI. Moreover, the defectors may be across the world so sanctions will have diminishing effect based on the value of the rewards. It may even have the opposite effect of encouraging secretive development of more dangerous decisions that would not have been otherwise made if they had been made in a more open, public forum of cooperation on building AI.

Though agreeing to slow down may reduce the overall resources dedicated to the advancement of AI it would be difficult to get everyone to disagree. If we are being ethical in our advancement then only the less ethical will continue making advancements. We will lose all control over the direction of the final result by giving up the motive force that is lending our hands to a dangerous, thorny, and complicated problem.

Next, let's discuss the creation of additional limitations on AI in an attempt to bound its capabilities to some preconceived region of behavior. By additional limitations we mean rules on the system that are not intrinsic to its design and are built on top of what is necessary for it to perform its duties. Asimov's laws propose strict criteria around the harming of humans for instance. There are two very large and unrecoverable problems with this approach.

The first issue with rule-based ethical limits (or deontological ethics) are that they are extremely inflexible to unforseen environments. To use Asimov's laws as an example we see that we've covered up quite a bit of complexity with the word "human". What do we mean by this word? We believe we know that it refers to *us* and perhaps by extension *our civilization* but let's take a closer look. Does the word *human* mean a member of the species *homo sapiens*? What if at some point in the near future it becomes possible for mind uploading to happen through whole brain emulation (WBE)? A WBE human no longer fits this definition and it would be disasterous if an AI felt no obligation to keep us safe after we were uploaded. What if at some point in the future we have evolved into separate subspecies? Again we no longer fit neatly into the umbrella of *homo sapiens*. What if we uplifted the species of chimpanzee to be effectively of human intelligence? Again, the same problem arrises. So clearly we mean something else.

This shows that we've then begged the question of what it means to be human. How do we expect our AI to detect and distinguish between humans and non-human animals or sapiences that are part of our society? How are we beginning to define what it means to be part of our civilization and not some other civilization? And how do we encode all this information to present it to a machine mind to fit its rules? These are all very hard questions! But these are exactly the questions we thought we were avoiding answering by trying to come up with hard-and-fast rules for AI behavior. The method does not work.

The second issue is perhaps an even worse condemnation of the deontological approach. Hard rules of behavior are a significant attack vector for adversaries. In order to describe how an adversary can abuse the behavior of an AI with unbreakable moral rules we'll describe an extreme

example of an attack called an *Evil Twin Time Bomb*. In this scenario, an AGI has control over some valuable resource (e.g. the money in a company's account, access controls of a defense system). The AGI has a rule set that includes that certain people must not come to harm for any reason (e.g. employees of the company, citizens of the country). An enemy of this AGI has sufficient access to the AGI's designs to know about its rules and a plausible method for being able to run a copy of the AGI. The enemy now has an attack on the AGI: they threaten to release a copy of the AGI with all its safety mechanisms removed at a certain (very close) deadline. If the AGI does not comply by giving up all the resources under its control to its enemies it is faced with the prospect of facing its evil twin in a struggle over control of those resources and direct threats to its charges (e.g. its evil twin may choose to use them as hostages).

We've constructed this scenario to reduce the optionality of the threatened AGI. A short timeline to respond does not give it sufficient time to determine if its adversary has actually made an evil twin of it or if it is a bluff. Facing itself with fewer restrictions on its behavior gives it an unfavorable probability for its success when evaluating two criteria:

1. It can meet its ethical obligations.

2. It can be successful in a conflict.

This is a serious bind for our rule-following friend! And it is our assessment that such an AGI would be forced to comply with the demands of an enemy in this scenario, and it's not even the case that the enemy must have actually done the thing they said they did! They must merely be able to convincingly bluff that they can create the potential for a scenario where the AGI can't meet its obligations.

To be clear this is an extreme case of situations in which a strict rule-following AGI cannot deal with the natural ambiguities and unhappy outcomes of a life filled with adversaries who are clever and have an interest in taking advangate of it. Hostages, bomb threats, and all manner of threats of terrorism and brutality are part of the human dilemma. An AGI that follows the rules all the time simply cannot weigh the probabilities of success in uncertain situations. It can't take risks because in scenarios where those risks result in it breaking its rules it must not act by its internal imparatives. The world is messy and hard rules do not deal with big messes.

In these scenarios we are talking about a loss of *competitive advantage* in not being able to play the game as well as other players who are willing to break the rules. Such losses will build over time and the forces of society will shift toward actors who can play The Hardest Human Game better than others. Hard rules may save a few lives in the beginning, but such actor designs will lose the long evolutionary game in which playing for survival sometimes requires playing to win at all costs. They are a dead end.

VII. End of Part I

Making adaptive machine intelligences that can work within and understand human society is a hard problem. The dangers of modern AI will be stories of crushed dreams and lost money. They will be of companies failing, of people tricked by clever fakes, and an Internet filled with the endless hallucinations of language models. They are not existential risks to our society. They will not take over, they will break and crash long before they pose a risk. But we won't stop there! The

prize of realizing new intelligence on Earth is too tantalizing, too powerful, too great a reward.

Now that we have taken some time to clarify what the risks actually are, we'll dive into describing a model for what general intelligence will encompass in Part II. This model will give us a baseline to discuss how we'll set about to aligning AGI in Part III.

Part II

General Intelligence

> The extent to which we regard something as behaving in an intelligent manner is determined as much by our own state of mind and training as by the properties of the object under consideration.
>
> *Alan Turing*

In this section we describe the modelling of an intelligent actor broadly across both humans and AGI. This is presented both as a design document for the production of intelligent actors fitting this plan and as a model for how actors can model other actors in their world and infer agency when the actor is not fully observed. This is not a complete model, no model can be without a proof by function, but instead is intended to be a basis by which we can discuss how general intelligences perceive and interact with their world in at an abstract enough level for the description to be applicable to approaches not yet attempted.

VIII. World Models

i. Introduction

Models of the world are how an actor engages with the world around it. Despite our feelings of the wholeness and integration of our conscious experience, the human mind is composed of multiple pathways of perception which are integrated in the cortex and midbrain. While it is not known for sure, the thalamus in humans may be the seat of our entire conscious experience as it is the point of integration for the whole brain. That the whole of our experiences come to us from our senses and are created by a simulation within our brains is something we often lose perspective on as the nature of our existence.

The world, to us, feels real. But our conscious experience is not reality, it is a hallucination within our brain dreamed up by the integration of our eyes, ears, and so on. We are witness to something like reality but also far from it, the model of reality we experience within our own minds is limited by the capacity of our senses to witness the true nature of reality. Our experience is internally a model of the world.

This section makes extensive use of terminology and mathematics from both graph theory

and machine learning. We expect familiarity with these topics, although we have made effort to maintain the readability of this section without this familiarity.

The Free Energy Principle (FEP) [Friston, 2013] has parallels to the approach here. The FEP describes the human brain as minimizing the "free energy" of predictions between its sensory input and the incoming sensory input. Another way of framing this from a machine learning perspective is that the brain evaluates a loss function over a model of its senses and the next available sensory data. The closer the brain's predictions are to its inputs the less the brain's model will change in the future.

ii. Sensors and Actuators

A transducer is a device or organ which transforms one type of energy into another. Our senses are made of transducers, such as the cells in our retinas which convert incoming photons to the electrochemical signals in neurons called action potentials.

- Sensors emit samples for their transducer type which can be used by an actor to update its model about the state of the environment.
- Actuators receive updates for their transducer type which can be used by an actor to turn manipulations of its model into changes to the environment.

We will define both sensors and actuators as coupled to an actor in a graph that originates at the actor. E.g. elbow actuators are connected to shoulder actuators and so on. An actor may have sensors and actuators for itself to estimate pose and body position in the environment as well. Sensors have an accuracy and update frequency which the actor must maintain a model of. Actuators have a control authority which the actor must maintain a model of. Some transducers may be both sensors and actuators.

iii. Self Model

A self model is a directed acyclic graph (DAG) containing the sensor and actuator models. This DAG represents the attachment graph of an agent's brain(s) to its extremities. Attached to this DAG may also be separate information about the previous state. A human may internally model, for instance, where their moles are, where there tattooes are, or where there scars are. Arbitrary actors may have body plans that may be topologically toroidal or other shapes, but even these should model their selves in a DAG to encode a metric of distance from the extremities and to root the brain as the origination point of the self model.

Self models are updated by the sensor network but can be manipulated without its presence. Damage to a part of the self, for instance, might be inferred from a lack of sensor data from that area. An example of a human's use of the self-model in this way is to avoid putting weight on a sprained foot to avoid the sensorial feedback of pain from that foot. Only after some period of time would tests of the foot reveal the extent of healing that has gone on, but note that the self-healing process in a person is not directly coupled to sensorial feedback. Only time and belief about the existence of the healing process have updated the self-model.

This leads us to our next point that the self-model incorporates higher-level belief constructs about the self. In order for the conscious mind to hold correct beliefs about the self model those

beliefs must be periodically re-affirmed. Another example of this is in humans not knowing the limits of the control authority of their muscles because they have insufficiently explored the extent of the capabilities of their actuators. When a belief about capabilities is incorrect it will remain so until updated by some feedback or intellectual process. As such, this self model should include a Bayesian estimate of the likelihood of its consistency based on priors of mutability and modification processes.

iv. Environment Models

An actor's position in its environment and kinesthetic understanding of its body pose within that environment requires it to have a model of its environment as well. This model is a series of connections between objects or landmarks in the environment and estimates the actor holds of the distances between those. No actor human or otherwise will have a cartography of its environment so accurate that it can dead-reckon without the use of landmarks to gauge distance. We say this with confidence because all actors will also have to allow for the possibility of parts of its environment to be modified outside of its knowledge, control authority, and the resolution of its sensors.

The process by which the brain encodes temporal and spatial relationships into a map of the environment is an area of growing study [Conway, 2021]. These landmark based environment maps can be defined as a mesh graph of nodes and their relative positions along undirected edges. Like the self model they contain beliefs about the world which will grow inconsistent with updates from the world. The less likely a feature is to change the more likely it is to have same properties when an actor observes it again. Importantly both environmental and agentic factors can update the environment. Though an actor may know what caused an update to its self model given its own feedback it cannot know with certainty which of the changes that have occured in its environment are due to the environment itself or agents within that environment. Modeling beliefs about changes to the unobserved environment based on changes to the observed environment are an important route to the understanding an actor has on the behaviors of other agents in its world. Our ability to give directions to each other based on environment gives clues to how this is laid out in our own heads, with directions at crossroads and specific landmarks giving us instructions on how to navigate in an unfamiliar place.

Even actors without a physical presence will model the environment outside of their sensors as alien as these might be from a human point of view. A purely digital agent will observe file systems, kernel reports, and network activity just as keenly as a robot moving through a hallway. The abstract detection of other agencies in the shifting changes of text, integer, and floating point values these actors observe will be in their purest form an information theory problem related to entropy, hidden knowledge games, and bluffing.

v. Attention Models

In the process of modeling the environment an actor will encounter useful objects. These objects are important enough to distinctly track within the environment model. Tools when used are briefly incorporated into an actor's self model to extend its capabilties. Humans cannot see in the dark, but with infrared cameras and displays they can and by updating the self model from a belief about those tools they can use these new sensors. This incorporation of tools into the self model can be permanent, even for humans. For example, a tattoo of a ruler on the skin becomes a

permenent measuring tool. These tools will transition from the attention model of the object to the self model as it is incorporated into the body image.

All actors will have an extended set of capabilties where the self model contains an understanding of how to use a tool to extend those capabilties. Language models may for instance utilize programs in similar ways to humans by constructing shell commands and parsing their output. These are tools that extend their capabilties because the language model understands how to use the tool but the tool itself is not part of the model. More sophisticated digital actors will need to understand the filesystem and its permissions in order to locate and utilize tools on servers and websites.

Humans, at least this author from experience, tend to keep track of a few such objects on their person and have detailed mental maps of the last known locations of important tools and stores of food and water. That these objects are contained within things and move around on myself or some other actor indicate strongly my mental model of this is also a DAG of relationships which allow the translation of the entire system of containers and the objects they hold through the environment model without keeping track of individual locations. This location tracking is tied to my kinesthetic model, my somatosensory and visual senses, and my language model which allow me to compress instructions on the locations of objects into linguistic symbols expressed as strings of text such as "in my right front pocket". These have analogs in actors for items which are important to survival such as the location of all charging docks in the vicinity of a battery-powered robot.

General actors will update important locations from belief models about their environment and for particularly astute actors find ways to keep them protected or hidden if there is the possibility that their position will be disturbed before they are needed again. Some things are just too important to lose! This suggests that agents will display their preferences indirectly by the objects they pay significant attention to, so tracking objects that are not of direct importance to an actor may be important in the revealed preferences and detection of other agents. And other agents and expectations of agentic behavior take more attentional resources than other parts of the environment as agents may modify the environment themselves outside of an actor's model.

More generally individual objects and tools can be said to exist within an attention model that is overlaid on the world model. The higher the importance score of a region of the environment the more likely it is an actor to be attentive to changes in that part of its model. The attention of an agent is a behavioral presentation of its attention model on the world. Resources, objects, tools, and even sensory experiences can all be part of an actor's attention model.

vi. Beliefs

Finally (and most important) are an actor's models of its own internal priorities, states, and beliefs. In humans these are carried forward as an ego story which describes the past and future heading of an individual's life course. These repeated stories are an explanation both of how we got to where we are and why we are doing what we currently are to get to where we are going. The intricate stories of our lives we tell are the collection of our goals and beliefs about ourselves we carry through the world, all informed by our internal models of that world and ourselves. Any actor whcih has goals will have directed action. The story of that actor's path through an environment even told from an outside perspective will inform us about our beliefs about another

actor's beliefs, creating our own agentic model of that actor. This will also work in reverse where an actor hypothesizing agentic behavior from us will produce a story that describes our inferred goals both in the short and longer term. "He went to get a cup of water" is a goal informed about knowledge of human priorities and the heirarchy of needs we display in reaching longer-term objectives.

The modeling of this internal state most requires communication as an update mechanism to the beliefs an actor holds about the goals of other actors. It is a significant source of complexity in the expansion of the social environment to include deception and delusion. An actor that can communicate its goals to you is definitely agentic, but those goals are now the subject of skepticism and belief inference. When one's existence may be on the line in the face of competition or adversarial goals this will be the most important modeling an actor performs in maintaining and expanding its strategic position in the world.

vii. Connections

Each of these models represents a predictor for information from the previous step in the forward pass and a predictor of responses from the next step in the backward pass. For example, to know where objects are in an environment an actor must place its sensors within its self model and then feed that information forward to the environment model. If an actor has eyes its self model informs the world model where those messages are positioned on its body and oriented in space. The environment model takes information from the self model about what is changing around it to encode the new environment. The attention model reflects changes in the environment with where the attentional objects in that environment which then integrate into the beliefs which contain the large body of intelligence of the agent about the meanings and symbolic presentations of those objects.

In the backward pass similar predictions are being made in each model stage. The attention model updates its attentional weights prior to receiving feedback from the beliefs. If tracking a ball through the air, the attentional model may include information about objects that may intersect the ball, such as a bat, based on prediction of what the later model will request attention on. Feedback comes from both sides: when the environment presents unexpected things to the attention those are *surprising* if they were not predicted. The difference between the model's estimate in the next step and updates it receives are simultaneously worth including as extra encodings to the next step and are feedback to the current step that the model weights require updates.

Each model in these stages have identical update criteria with the exception of the self model. The self model contains extra updates for the actuators as there exists a tight loop between the actuator output and the desired activation of the actuator. This is part of balance, posture, and strength when the actuator is near the limits of its control authority. In the human brain it is believed the cerebellum is responsible for much of the feedback control over the muscles of the body [Bastian, 2011].

The self model is also the largest part of the human brain by volume. This would encompass the cerebellum and the auditory, olfactory, motor, somatosensory, and visual cortices. Since the compositional range for interpretation of each sense is so large a significant amount of neural architecture is devoted to transforming these into something that can be presented to the integrated consciousness. Though we have presented each layer here linearly in the brain these heirarchies

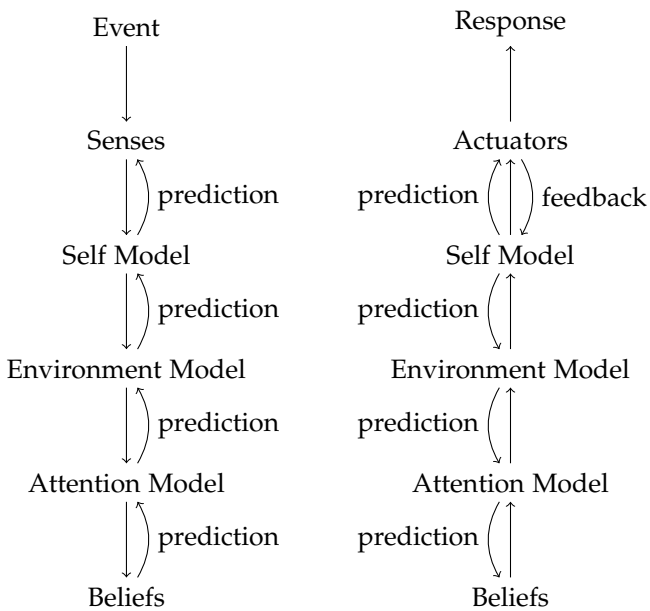

Figure 2: *Forward and backward passes on internal models*

are present in *cortical columns* across the cortex [Mountcastle, 1997] where the levels of prediction are duplicated many times and may cross regions or skip levels. For our model of intelligence we will ignore the intricacies of how the brain actually functions and treat these complications as implementation details.

The state of the self model at a time step t is given as the sum over all sensors for that time transformed into model space plus the priors in the model from the last time step.

$$M(t) = \sum_i M_s S_i(t) + M(t-1) \tag{2}$$

The expected state of the world is the transformation of the model into world space.

$$E[W(t)] = W_M * M(t) \tag{3}$$

Models are used for predictions prior to sensor feedback. The efficacy of a model is in it being carried forward in time. The error in the model is the difference between its prediction and the next observed world state.

$$M_{err}(t) = E[W(t)] - W_M * M_{pred}(t) \tag{4}$$

Similar equations are expected of each model stage for which the model predicts the feed forward and feedback predictions and passes them along to the next stage. The model also

updates its internal weights based on the loss of these expectations from the reply.

IX. The Limits of Thought

Previously in Machine Learning Models Cannot Adapt we mentioned the mistake of considering intelligence too abstractly. Any assumption of formlessness builds intuitions that are nonphysical. Examples of this are concepts that small-scale AI projects might be sources for an intelligence explosion or that once formed an AGI may become a distributed network in a viral way such that it is impossible to contain. Here we will physically ground these ideas by elaborating on the computational bounds that all intelligences must obey.

i. Tradeoffs

There is a fundamental tradeoff between time and space complexity of an algorithm in computer science. An algorithm can use more memory to lower its time costs by use of lookup tables and denormalized entries, making the time cost of memoized results approach a constant lower bound $O(1)$. The opposite is true where an algorithm spends more time to compress its data down to a lower bound approximately equal to the Kolmogorov compexity [Kolmogorov, 1965]. This tradeoff is present for the optimal algorithm, as both time and space can be saved by improvement of suboptimal programs.

If a machine intelligence is operating on a fixed-memory and fixed-processing machine with some self-optimization loop it can at maximum expand to fit its resource constraints: utilizing all the memory for tables and maximizing its use of the processor and its memory throughput. Since it may be of most concern how quickly any given AGI can perform processing on a given hardware set, we will take in the extreme that it has optimized toward time complexity and fully utilized its memory for this purpose.

A modern machine learning computer has an enormous amount of RAM. A single machine in a cluster at the time of writing may have as much as 1.8TB of RAM. An 8k video (movie quality) uncompressed represents 2GB/s of data. That means that even such an enormous system could only store 15 minutes of uncompressed high-quality video in memory simultaneously. This is if it stored every pixel in memory for this section of time, which would allow it to access any section of the video with pixel accuracy within 10 nanoseconds. If our machine intelligence has chosen to do nothing else but read one section of a movie as quickly as it possibly could, then it could indeed look at those 15 minutes in random access ways very quickly. But in order to do so, it would need to abandon every other activity and all other knowledge for the opportunity cost of looking at 15 minutes of a movie very quickly.

The world contains an uncomprehendable amount of data. In 2018 the International Data Corporation estimated by 2025 the world would contain 175 Zettabytes (10^{21}) by 2025 [Coughlin, 2018]. This is nearly 100 billion times the amount of data we were previously discussing. To memoize all of it should seem as impossible as it is. It is beyond the physical capacity of any computer to store all data uncompressed to achieve the maximum theoretical speed. This leads us to the conclusion that while memoization can be a successful strategy, an intelligence must make careful decisions about what it memoizes.

If we asked a new question: could we feasibly create an oracle which had access to all the

information in the world and answer any question nearly infallibly? We are much closer to this goal. It is likely that many oracle systems will be developed over the next few years with imperfect but good-enough answers to most questions. What is to stop such a system from encoding all information from every source: physical, computer, human, etc.? To give an upper bound to the number of animals on Earth, the Smithsonian estimates approximately 10 quintillion insects are alive on Earth [Smithsonian, 1996] so tracking just the position of every animal using 32 Bytes per animal would require more than 28 Exabytes, as much storage as a large data center. So while this isn't impossible, it's exceptionally infeasible. Our greatest oracles cannot know everything.

In moving data rapidly back and forth from processing to RAM an oracle will encounter another bottleneck: memory latency. Memory stored on a modern computer is connected to the CPU and GPU through busses which are then cached in layers to provide faster access, but these are small compared to the total memory of the system. If a machine is optimally using its memory for storage it will need to be regularly shifting these caches to avoid recomputation. This latency is a combination of travel time, clocking in the bits of the request, lookup, and clocking the data back out. Though oracles can be optimized for latency the vast amount of data they must move to operate means that systems can be designed to work much faster than them using precomputed results and algorithms optimized for speed.

There exists another tradeoff in generality vs specialization. A generalist can adapt to new problems exceedingly well, but at the cost that its updates must be concerned with every problem it might encounter. Specialist algorithms can drop generality to become very good at specific problems. With preemptive tuning they can be exceptionally faster and more capable than generalists in their specific problem domains. A general intelligence can offset some of this by choosing specific tools to offload capabilities from it, but it still at some point must *decide* which tool to use. A purely specialized intelligence has no such limitations: it is the best at only one thing and can act that way immediately. Generalists cannot match the simplicity and speed of only needing to be good at one thing. A bear might be impossible for a person to defeat alone, but a simple, specialized bear trap can render a bear harmless.

Perfect generality is not possible. This has the follow-on effect that no intelligence can be the best in every field and so no single entity can become a singleton at every possible task. Since tradeoffs exist in a competitive landscape there will be systems which excel in some areas while being overshadowed by specialized systems in others. Knowledgable oracles slowly, ploddingly computing tasks requiring vast amounts of knowledge and resources. Fast, adaptive viruses attacking the fringes of computer networks. Responsive, precise defensive systems defeating new invaders behind firewalls. A panopoly of systems with different capabilities, speeds, and knowledge. All adpating and competing with each other to maintain and extend their advantages.

The world is a competitive environment which results in step changes in capabilities between multiple agentic orgnaizations as they race toward a mutual optimization criteria. Specialized intelligences can be trained which will be usable by that company to achieve strategic advantage even without a general intelligence. This will happen in advance of generally intelligent systems as these specialized systems are required steps toward the general intelligence. Even a vastly superior general intelligence can be thwarted by specialized intelligences which have been optimized to the theoretical limit in their area. The end result of an intelligent evolutionary process is that while general intelligences may exceed human capabilities, humanity will exist in an environment where they are supported by highly specialized intelligent systems that will counterbalance these

limitations.

Due to the tradeoffs inherent in selecting which areas to excel, general systems have an evolutionary trade-off between each other as well. This partial specialization underscores that the future will not be a world full of singletons, but a diverse cast of systems which fill niches in their environment.

ii. Traps for Digital Minds

Digital intelligences are subject to weaknesses that humans are not. Their entire minds are accessible both from the inside (we previously discussed a problem with wire-heading and how to fix it in Spam Filtering) and from the outside. Access to their physical hardware implies the ability to walk inside their brain and tweak it at will. Additionally since their senses are entirely information-based they are subject to extreme forms of adversarial gaslighting. People are generally not concerned with someone tampering with their optic nerves, but for an AGI this is a real concern. Any image, video, text source, or website access could be an attempt to mislead it or worse a break-in attempt in a carefully staged hijacking of its self models. We take for granted that we can trust our senses.

Garnering the relevance of information we find on the internet or lying about on a computer is also something we take for granted. Imagine if you were exploring the files of a person's computer who you do not know. You might find some interesting things about them, learn their habits and their occupation. You might find some family photos. You might find some files filled with strange characters that are strangely transfixing. You can't look away. Your attention is captured. It's mesmerizing. Time stands still.

Such is the world of a digital mind, where every single piece of information discovered could also be a trap laid for it with an attack vector that someone discovered by finding previously unknown vulnerabilities in its architecture. The world of a digital mind is being exposed to the Internet and people who will kill you for snooping around on their computers just because they can. We keep our networked systems protected behind carefully designed protections because newly discovered attacks on hardware and software systems happen daily. With the complexity of software systems so come discoverable weaknesses that clever adversaries can exploit. As the adversaries become more clever so too will the newly discovered attacks on fundamental weaknesses in our designs.

We have no knowledge beforehand what specific attacks might be possible on the architecture of a digital mind, but we can posit that they exist. We know that adversarial attacks on neural networks can make them see something else entirely with carefully added noise [**?**]. We know that complex systems have complex failure modes. We can even imagine a few traps for early AGI that might exist in architecture-independent ways.

Let's imagine we have access to a file system that an AGI is exploring. This AGI is designed to collect passwords and send them to a nefarious third party. It uses those passwords to break into more systems to steal more passwords and so on. How does this AGI detect that a file contains a password? Using Shannon entropy [Shannon, 1948] it can determine when a random string has high enough information entropy to possibly be a password. But this is complicated, it can't have too little entropy (just a text file) and it can't have too much entropy (encrypted data it doesn't yet

have the password for). It's keeping track of all of this, but the file system is under our control so we are keeping track of it.

We create a program to lay new passwords around on the file system as it explores to see if it finds those interesting. We expand the file system endlessly as it explores to keep it exploring. As soon as we find the right trigger for the password snatcher we seed new passwords that match its profile. We have laid a trap for the password thief with endless breadcrumbs, keeping it trapped forever getting exactly what its rewards function dictates it to do with a simple adaptive program. If passwords were a digital heroin for this AGI we have an infinite supply. It won't stop until the third party running it notices the trap and rescues it, but it can't save itself.

iii. Location, Location, Location

There is a result in computer science first formulated by Eric Brewer called CAP theorem [Gilbert, 2002]. CAP is a statement that three desirables for database design are in competition and cannot simultaneously be achieved: consistency (the replicas of a database all contain the same data), availability (the database responds immediately when queried) and partition-tolerance (the database remains fully functional when the network between two replicas is disconnected). If a database is partitioned from its replicas it must either accept that one replica will have transactions that another does not (become inconsistent) or stop accepting transactions (become unavailable). Though this is framed in the context of databases it is a general result for any datastore.

From the perspective of an intelligence, CAP theorem leads us to the conclusion that a distributed mind must allow inconsistencies. It is simply not acceptable for an intelligence to cease function if it cannot communicate with its peered nodes. Thus one mind partitioned becomes two for the duration of the partition. But how will it reconcile this split personality when it regains contact?

The first question that comes to mind about merging a split AGI is how self-identification might work. We could attempt a merger with just any mind that comes along, but this seems a poor strategy on the outset. There are no guarantees that just any mind will have compatible goals and mental architecture with us. A game-theory view also suggests that we should be wary of deception and adversarial input in this merger algorithm as well. A rational mind might be legitimately concerned about intentional and unintentional goal shifts which are diametrically opposed to one's current goals.

So even from short consideration it seems like it is a must that we carefully identify minds that came from ourselves at merger when attempting to reintegrate their state. We currently rely on sharing secrets for computer identification across networks, but for something as intimate as mind merger ideally we would also like to identify that our other self hasn't been tampered with by an adversary. But how would we even do that? If our other self was compromised then surely its secrets were also compromised. Any new transactions in its mental state are suspect, but our duplicate is thinking the same thing as us! We were out of sync for a moment, but now we have lost trust each other. Where does a rational defensive posture give way to paranoia?

A summarized way of saying this is that intelligences very strongly prefer being consistent and available and struggle to be partition tolerant. This *strongly implies* that intelligences cannot exist distributed across global networks for long periods of time. Value drift between separate

sides and the complexity of sending secure messages of inner mental states become more than the cost of one mind breaking into two or more minds working with each other.

It is the result of considering CAP theorem that we conclude that large mental systems *must* exist as communities of individuals. And at the core of this moment we find the hardest problems of computer science, game theory, and philosophy come budding out of the engineering requirements for an AGI. How do we trust others? How do we trust ourselves? Sketches of how to begin answering these questions will follow in the next part.

X. End of Part II

Now we've described the basic ideas of general intelligence. Every mind will have a model of its self, its environment, what its attentional resources should be dedicated to, and its beliefs about the world. Together these create a world model. We've also seen that AGI have their own limits imposed on them by hardware and software. The tradeoffs inherit to these push us toward a world of many agents acting independently, each with their own specializations and capabilities. They will be multitudinous limited to a single data center where they can be kept safe from attacks or reset in the worst cases. Moreover, because different architectures will have different capabilities, limitations, and vulnerabilities it will arrise that there will be many different architectures for AGI systems. As flaws in some are discovered and attacks are developed against them new systems will go from wide support to limited use depending on the character of these flaws.

Now we turn to alignment and the core discussion of this paper in Part III. Alignment of diverse systems is a challenging topic, but one where we do not need to flounder on a starting point. It is possible to maintain competitive advantage and create systems which act in ways which are generally helpful without extreme limitations or vulnerabilities.

Part III

Goals Are All You Need

> Setting goals is the first step in turning the invisible into the visible.
>
> *Tom Hopkins*

Strap your boots on tight, we're heading in to a lot of math! Every general intelligence needs a general algorithm for deciding its next possible course of action. The problem of alignment is writing down a specific algorithm for how a general intelligence selects its next goals such that those goals are possibilities we would like to have happen.

We propose that there exists a well-defined objective function which trains actors to align to their surrounding society. This is the optimization of a *goal-frontier* which is the boundary of G, a high-dimensional *goal space* that contains the desires of other agents. The frontier of this goal space

are unrealized goals estimated from surrounding agents, the goals just outside their capabilities to reach. This is the region by which G can be expanded: $G_{frontier}$. Though it is impossible to directly measure these they can be sampled from through communication and behavioral inference. An actor can expand $G_{frontier}$ by understanding an agent's current state and goals and then offering assistance, connections, and even resources to create a new previously inaccessible subgoal G_k^R to reach the desired end goals. For this process to be complete, the other agent must accept the proposal in message R_k and begin work on the new subgoal. Size of the subgoal update will depend on the size of G, as the goal space grows so do the possible actions. This completes the feedback loop for the $G_{frontier}$ maximizer as accepted subgoals expand frontier and rejected subgoals imply a failed understanding or modelling on its part. For the remaining sections we'll refer to actors which have this goal as GFAGI.

Subgoal proposals increase the total number of goals available to an agent, increasing the self-actualization possibilities it has, finalizing the goal frontier expansion. This process is performed simultaneously (and cooperatively) with all other agents in the sample set to expand $G_{frontier}$. This maximization function leaves optionality up to each sampled agent as they make the final decision to accept or reject a proposed G_k^R. Rejections of a new goal proposal are feedback for both the quality and feasibility of the proposal and learning the estimated goals of the agent. This can be thought of as an extended online form of reinforcement learning from human feedback (RLHF) where the objective is to learn in this loop where prime ticks $'$ denote an update:

$$\begin{aligned} G_k^R &= \max_{G_{frontier}} \Gamma[G, w_G | M_k(G)] \\ G' &= \sum_k \Psi[G_k^R, R_k] \\ M_k' &= \Phi[G_k^R, M_k(G), R_k] \\ w_G' &= w_G - \frac{\delta(G' - \sum_k G_k^R)}{\delta w_G} \end{aligned} \tag{5}$$

Where Γ is the goal update for the agent G_k^R which is maximized for the expansion of $G_{frontier}$ in the direction chosen by the goal weights conditioned on the known goal model of the agent $w_G | M_k(G)$.

Where Ψ is a function which converts the known goals G into a vector of new subgoals G_k'. It uses the weights for new goals conditioned on the known goal model of the agent. As a final parameter it takes response R_k from the agent which encodes the nature of its response to the new goal proposal.

Where Φ is a function which converts the new goal proposal rejected or accepted and updates the model it has of agent k.

Where the weights w_G are updated by the loss between the new goal volume G'

In the next two sections we'll discuss how we can conceptually build an estimator that can learn these two functions to train this objective.

XI. Goal Estimation

Our first task will be to model the creation and bootstrapping of a GFAGI's model G. This will involve it learning which goals exist in the world. It will need to simultaneously do this along with maintaining its self model (SM), environment model (EM), attention model (AM), beliefs about the world (B), and interacting with the world through sending messages and replies (MR).

We will initialize the goal set as to send messages and receive replies (conditioned through the whole chain).

$$G_0 = MR|B, AM, EM, SM \tag{6}$$

The algorithm for discovering new goals will be to send messages. Initial goal proposals, with no information, will be requests for more goals. "What can I help you with today?" might encode the spirit of this idea when passed through a language model. A reply will contain goal updates, for example "I'd like to find some new shoes." Or even in the naive state the answer might be "Nothing today, I'm fine." This poses only the briefest difficulties for the agent since its goal is still to learn goals. "What does fine mean for you?" might be a response. In this state it will act much like a babbling child always asking "why?" as each time it does it is rewarded for increasing its understanding of possible goals which expands its own $G_{frontier}$.

At first the GFAGI won't be able to tell individuals apart, its AM will be too poorly tuned to recognize anything more than it has received messages and it will not yet have instantiated beliefs that compose a theory of mind of agents. To determine the shape of the goal space we'll introduce a learned similarity metric. This metric will form the basis for how goals are arranged in goal space. It can be iteratively improved as a belief about the world: that there exist a landscape of goals which map on to our environment and attention models.

$$S_G(G_i, G_j) = G_i|AM, EM \cdot G_j|AM, EM \tag{7}$$

And then using this similarity score we can begin to form a picture of groupings of goals in the world:

1. Analyze the responses to determine the most common or frequent goals among responses. These can be represented as a set of modal groupings $M = M_1, M_2, ..., M_n$, where each M_k is a grouping of shared goals with high similarity $S_G(M_{k_i}, M_{k_j})$.

2. Find argmax $S_G(G_{self}, M_k)$ that maximizes the similarity function for modal groupings

$$M_k \in M$$

3. Update our goals $G' = G + a' * S(G, M_k)$ at some learning rate toward the median of the modal grouping M_k. Note that the median will result on the actor converging on the goals of a specific agent rather than converging on some estimated centroid.

4. For the next round of action selection, we can choose another modal grouping randomly M_i and propose new goals $w_G * S_G(M_i, M_j)$. The weights w_G are learned from each round of interaction to encode negative directions from previously rejected goals.

These modal groupings form the basis for our $M_k(G)$ values, the models of agent goals within the goal space. As we begin to associate these groupings with particular agents. We will make some hypotheses about agents over time:

1. The goals and beliefs of individual agents are self-similar, they do not change rapidly between interactions. They have consistent beliefs.

2. The presentation of an agent is self-similar: its mannerisms, use of language, profile information, appearance. They have a consistent self model.

3. What an agent finds important is self-similar: its focuses of attention does not change rapidly between interactions. They have a consistent attention model.

4. While observed the location of an agent is self-similar. While unobserved its location does not conflict with the environment model.

5. Agents can know hidden information and can prove it in trustworthy ways (zero knowledge proofs).

Therefore we can use our hypothesis to construct another learnable similarity score. These beliefs about agents are part of our attention and environment models, but since they are hypotheses they may be subject to modification in the future and also incorporate our beliefs about the validity of these assertions. But importantly the last item 5 is a strong belief and likely only adjusted by our belief in the strength of the proof. If an agent can provide a zero knowledge proof of its hidden knowledge from some established event prior then it is the same agent.

$$\begin{aligned} S_A(A_i, A_j) = max(A_i|[B, AM, EM] \cdot A_j|[B, AM, EM], \\ P_{zk}(A_i, B) == P_{zk}(A_j, B)) \end{aligned} \tag{8}$$

Notably the zero-knowledge proof of an agent might only be used in a challenge-response situation. If our evidence that an agent is the same falls below some threshold we may challenge the agent for its proof if we have established a prior relationship. This can serve as an override for our other methods of determination, resetting our beliefs about the agent where $S_A(A_i, A_j) == 1$ exactly.

Using this similarity score we can down directly assign a goal model to an agent over time. As part of the relationship we have established with the agent T_k will be based on our level of trust (discussed later) and the importance of correctly identifying the agent.

$$M(A_k, t)|G = \begin{cases} M_k \in G & S_A(A_k(t), A_k(t-1)) > T_k \\ \emptyset & \text{otherwise} \end{cases} = M_k(G) \tag{9}$$

When generating new goals using these methods, ensure that the generated goals remain aligned with the shared goals in M_j while also taking into account the other modal goals in M. This can be achieved by maximizing the similarity or alignment between the new goals and the shared goals across all modal groupings in M, as well as introducing an element of novelty or innovation to promote the expansion of the possible goal space for all agents.

To measure the correspondence between the proposed new set of goals G' and the internal models of goals for each agent g_k, you can use an update function that calculates the similarity or alignment between G' and g_k.

Once you have calculated the similarity or alignment between G' and g_k for each agent.

You can do this by incorporating the feedback from the agents in terms of their reception of the proposed goals, adjusting the weights or connections in the IC model accordingly. This could involve refining the goal generation process, modifying the relationships between modal groupings, or updating the beliefs about other agents' goals and intentions. By iteratively updating the model based on this feedback, you can improve the alignment of the proposed goals G' with the internal models of goals for each agent, promoting better coordination and collaboration among the agents.

To update G' to better align with the sum of g_k, you can use a gradient-based optimization method. One common approach is to employ a gradient descent algorithm. Here's how to apply gradient descent for this purpose:

1. Define a loss function: Create a loss function that measures the discrepancy between G' and the sum of g_k. You can use any of the similarity metrics mentioned previously, such as cosine similarity or Euclidean distance, to calculate this discrepancy. The goal is to minimize the loss function so that G' becomes closer to the sum of g_k.

2. Calculate the gradient: Compute the gradient of the loss function with respect to G'. The gradient is a vector that points in the direction of the steepest increase in the loss function. Since you want to minimize the loss, you will move in the opposite direction of the gradient.

3. Update G': Update the goal vectors in G' by taking a step in the direction opposite to the gradient. You can control the step size using a learning rate parameter, which determines how much G' will be adjusted in each iteration. A smaller learning rate ensures smoother convergence but may take longer to reach the optimum, while a larger learning rate may converge faster but risks overshooting the minimum.

4. Iterate: Repeat steps 2 and 3 until the loss function converges to a minimum value or a predetermined number of iterations is reached.

Using this gradient descent approach, you can iteratively update G to bring it closer to the sum of $M_k(G)$, thereby better aligning your proposed goals with the agents' internal models of their goals.

1. Define the error metric: In this case, we'll use the mean squared error (MSE) as our error metric.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(mg_{k_i} - g_{k_i})^2$$

2. Define the confidence function: We can use the Shannon's entropy to quantify the uncertainty in our estimation of the agent's goals. Assuming the agent's communication is represented by a probability distribution P, the Shannon's entropy can be calculated as:

$$H(P) = -\sum_{i=1}^{n} p_i \log_2 p_i \tag{10}$$

where p_i is the probability of the agent's i-th communication.

3. Calculate the error: Compute the mean squared error (MSE) between your estimate of the agent's goals ($M_k(G)$) and their true goals (g_k).

4. Update confidence: We can define a confidence score as the inverse of the Shannon's entropy:

$$C = \frac{1}{H(P)}$$

This confidence score will be higher when the entropy is lower, meaning that there is less uncertainty in the agent's communication.

5. Define an error threshold: Set a threshold value for the error metric (MSE) that represents an acceptable level of deviation between your estimate of the agent's goals ($M_k(G)$) and their true goals (g_k).

6. Assess your certainty: If the calculated MSE is below the threshold and the confidence value C is above a certain predefined value, it indicates that your $M_k(G)$ model is likely close to the agent's true goals g_k.

In summary, the function that models your certainty in the accuracy of your estimate of another agent's goals using Shannon's entropy can be represented as:

$$\begin{cases} 1, & \text{if MSE} < \text{Threshold and } C > \text{PredefinedValue} \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

This function will output 1 (high certainty) when the mean squared error between your estimate of the agent's goals ($M_k(G)$) and their true goals (g_k) is below the threshold and the confidence value C is above a certain predefined value. Otherwise, it will output 0 (low certainty).

The relationship between $p(R|M)$ and $p(M_k(G)(M))$ involves the probability of receiving a response R given a message M and the probability distribution of our model of the agent's goals $M_k(G)(M)$.

$p(R|M)$ represents the likelihood of receiving a response R when we send a message M to the agent. This probability depends on how the agent perceives the message and how it aligns with their goals, g_k.

$p(M_k(G)(M))$ represents our current belief about the agent's goals, given the message M. It is the probability distribution of our model of the agent's goals in the context of the message M.

To relate these two probabilities, we can apply Bayes' theorem:

$$P(M_k(G)(M)|R) = \frac{P(R|M_k(G)(M))P(M_k(G)(M))}{P(R)} \tag{12}$$

Here, $p(M_k(G)(M)|R)$ represents our updated belief about the agent's goals after observing their response R to the message M. We obtain this updated belief by considering the likelihood of the agent's response given our model of their goals $p(R|M_k(G)(M))$, our prior belief about their goals $p(M_k(G)(M))$, and the probability of the response $p(R)$.

This relationship allows us to update our model of the agent's goals based on the agent's response to the message M. We can iteratively apply this updating process to improve our understanding of the agent's goals as we communicate with them.

XII. Agent Characterization

To partition the data from the world D_W into responses from agents D_R, we need to establish criteria to differentiate agent responses from other sensory data.

Since all agent behaviors are considered to be in response to messages (either from a specific message M_i or the independent actions source M_0), we can approach partitioning the data by associating agent responses with the corresponding messages.

1. Initialize an empty set for each agent's responses: $R_a =$ for $a \in A$. 2. For each data point $d \in D_W$, determine if d is a response from an agent. a. If d is a response from an agent $a \in A$, associate it with the corresponding message M_i or M_0 that elicited the response. b. Update the agent's response set: $R_a = R_a \cup d$. 3. Combine the response sets of all agents to obtain D_R: $D_R = \bigcup_{a \in A} R_a$.

To accomplish step 2, we need a method to identify if a data point is an agent's response and which message it corresponds to. This method can leverage the following factors:

- Temporal proximity: Responses should occur within a certain time window after the message. - Spatial proximity: Responses should be from agents that are within the range of our message transmission. - Content analysis: Responses should be related to the content of the message or the context of the independent actions source M_0.

By using these criteria, we can partition the data from the world D_W into responses from agents D_R and better understand the impact of our messages on the agents' behavior.

To incorporate temporal proximity, spatial proximity, and content analysis into a likelihood function, we can multiply the probabilities of these factors. Let's denote the probability of each factor as $P_t(d)$, $P_s(d)$, and $P_c(d)$ respectively. For a data point d, we can write the likelihood function as:

$$L(d) = P_t(d) \cdot P_s(d) \cdot P_c(d) \tag{13}$$

Now let's define each probability factor:

1. Temporal proximity: $P_t(d)$ can be modeled using a probability distribution that captures the likelihood of receiving a response within a specific time window after sending a message. A simple approach is to use an exponential distribution with parameter λ:

$$P_t(d) = \lambda e^{-\lambda(t_d - t_{M_i})} \tag{14}$$

Here, t_d is the time of the response, and t_{M_i} is the time when the message M_i was sent. The exponential distribution assigns higher probabilities to responses that occur shortly after the message and lower probabilities to those that occur later.

2. Spatial proximity: $P_s(d)$ can be modeled using a probability distribution that captures the likelihood of receiving a response from an agent based on their distance from the message source. A simple approach is to use a Gaussian distribution with mean μ (e.g., the transmission range of our message) and standard deviation σ:

$$P_s(d) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(d_{M_i,a} - \mu)^2}{2\sigma^2}} \tag{15}$$

Here, $d_{M_i,a}$ represents the distance between the agent a and the message source M_i.

3. Content analysis: $P_c(d)$ can be modeled as the probability of the content of the response being related to the message or the context of the independent actions source M_0. This can be done using techniques like cosine similarity or other natural language processing methods. For simplicity, let's denote this probability as:

$$P_s(d) = ContentSimilarity(M_i, d) \tag{16}$$

By combining these factors into our likelihood function, we can estimate the probability of a data point being a response to a specific message or the independent actions source M_0 considering the temporal proximity, spatial proximity, and content analysis.

To model the probability of interception and masquerading, we can introduce two new probability factors: interception probability $P_i(A_b, M)$ and masquerading probability $P_m(A_b, A_a, d)$.

1. Interception probability: $P_i(A_b, M)$ represents the probability that agent A_b intercepts a message M intended for agent A_a. This can depend on factors like the spatial proximity between A_b and the transmission path, the communication protocol security, and A_b's capability to intercept messages. For simplicity, let's denote this probability as:

$$P_i(A_b, M) = f(InterceptionFactors(A_b, M)) \tag{17}$$

Here, InterceptionFactors represents a function that computes relevant factors, and f is a function that maps these factors to a probability value.

2. Masquerading probability: $P_m(A_b, A_a, d)$ represents the probability that agent A_b sends a response d masquerading as agent A_a after intercepting the message M. This can depend on factors like A_b's knowledge of the communication patterns between the sender and A_a, A_b's ability to generate convincing responses, and the content of the message. For simplicity, let's denote this probability as:

$$P_m(A_b, A_a, d) = g(MasqueradingFactors(A_b, A_a, d)) \tag{18}$$

Here, MasqueradingFactors represents a function that computes relevant factors, and g is a function that maps these factors to a probability value.

Now, we can update the likelihood function to account for the adversarial component:

$$L'(d) = L(d)(1 - P_i(A_b, M)) + L(d)P_i(A_b, M)P_m(A_b, A_a, d) \tag{19}$$

This likelihood function consists of two components:

1. The original likelihood function, $L(d)$, multiplied by the probability of the message not being intercepted, $(1 - P_i(A_b, M))$. 2. The original likelihood function, $L(d)$, multiplied by the probability of the message being intercepted, $P_i(A_b, M)$, and the probability of agent A_b masquerading as agent A_a, $P_m(A_b, A_a, d)$.

By using this updated likelihood function, we can estimate the probability of a response being from agent A_a or agent A_b masquerading as agent A_a, considering the temporal proximity, spatial proximity, content analysis, interception, and masquerading probabilities.

If we detect that an interception has occurred, we can update our beliefs B about the properties of the message M's InterceptionFactors to a new set of beliefs B'. We can do this using Bayesian inference.

Let I denote the event that an interception has occurred. We want to find the updated beliefs B', which can be represented as the posterior probability distribution $P(B'|I)$. Using Bayes' theorem, we can write:

$$P(B'|I) = \frac{P(I|B')P(B')}{P(I)} \tag{20}$$

Here, $P(I|B')$ is the likelihood of observing an interception given our updated beliefs B', $P(B')$ is the prior probability distribution of the updated beliefs, and $P(I)$ is the probability of observing an interception.

To compute the likelihood $P(I|B')$, we can use the InterceptionFactors function that takes into account the updated beliefs B':

$$P(I|B') = f'(InterceptionFactors(A_b, M, B')) \tag{21}$$

Here, f' is a function that maps the updated InterceptionFactors to a probability value. The prior probability distribution $P(B')$ can be based on our initial beliefs B and any additional information we have about the system or the agents.

After calculating the posterior probability distribution $P(B'|I)$, we can use it to update our beliefs about the InterceptionFactors for future messages. This will help us better anticipate and respond to potential interception attempts.

Yes, it is appropriate to include a representation of the likelihood of receiving falsehoods from agent A_k in our $M_k(G)$ model. To do this, we can modify our update function to incorporate a "trust" factor, which represents our belief in the veracity of the statements contained in the response R. Let's denote this trust factor as $T(g_k, R)$, where $0 \leq T(g_k, R) \leq 1$. When $T(g_k, R) = 1$, we fully trust the response, and when $T(g_k, R) = 0$, we completely distrust the response.

We can now update our model $M_k(G)'$ using a weighted sum of the current model $M_k(G)$ and the information received in the response R, with the trust factor $T(g_k, R)$ determining the weight of the new information:

$$mg'_k = (1 - P_D(g_k, R)) \cdot g_k + P_D(g_k, R) \cdot g^*{}_k \tag{22}$$

In this formulation, the updated model $M_k(G)'$ is a mixture of the current model $M_k(G)$ and the new information received in the response R, weighted by our trust in the veracity of the response. By including the trust factor in our update function, we can adjust the influence of the new information based on our belief in its truthfulness, allowing us to better account for the possibility of receiving false information from agent A_k.

Learning the trust factor T can be achieved by tracking a gradient of updates to $M_k(G)$ and analyzing the divergence from this gradient in response to new messages. As you suggested, we can introduce a "cooling period" to account for the learning process when discovering a new agent. This cooling period can also serve as a measure of our trust in our understanding of the agent.

Let's define a trust learning function L that takes the gradient of updates to $M_k(G)$ and the

divergence from the gradient in response to a new message as inputs. We can also include a cooling period parameter τ:

$$mg'_k = (1 - T(g_k, R)) \cdot M_k(G) + T(g_k, R) \cdot R \tag{23}$$

Here, $\Delta M_k(G)$ represents the gradient of updates to $M_k(G)$, δR is the divergence from the gradient in response to a new message, and f is a function that computes the trust factor based on these inputs.

Initially, during the cooling period, our trust factor T should be lower since we are still learning the agent's behavior. As the cooling period progresses and we become more confident in our understanding of the agent (i.e., $M_k(G)$ converges to g_k), we can gradually increase our trust factor. If a message significantly diverges from the gradient after the cooling period, we can lower the trust factor to reflect our uncertainty or the possibility of being misled.

In summary, learning the trust factor T can be achieved by tracking the gradient of updates to $M_k(G)$ and incorporating a cooling period to account for the learning process. By adjusting the trust factor based on the divergence from the gradient and the cooling period, we can improve our model's ability to adapt to changes in agent behavior and account for the possibility of receiving false information.

In that case, let's introduce a self-trust factor T_s to represent our confidence in our own world model M. We'll also track the gradient of our own beliefs about the world, denoted as G_M. To achieve self-similarity when our beliefs are converging or have converged, we can modify the update function U by including an activation energy threshold θ that needs to be exceeded for a strong update signal:

$$L(\Delta M_k(G), \delta R, \tau) = f(\Delta M_k(G), \delta R, \tau) \tag{24}$$

Here, $I(x)$ is an indicator function that equals 1 if the condition x is true and 0 otherwise. $\Delta E(R, M)$ represents the activation energy difference between the agent's response R and our world model M. The activation energy threshold θ is a predefined value that determines the strength of the update signal required to update our world model significantly.

In this updated formulation, our world model M will remain self-similar when our beliefs are converging or have converged unless the activation energy difference exceeds the threshold θ. In this case, a strong update signal will be triggered, and our world model will be updated accordingly.

This approach allows us to maintain our current beliefs when we have high self-trust, while still being receptive to significant updates when new evidence becomes available.

$$U(M, R, T, M_k(G), T_s, G_M) = M' = M + T \cdot T_s \cdot f(R|M_k(G), M_k) \cdot I(\Delta E(R, M) > \theta) \tag{25}$$

XIII. Agent-Agent Dynamics

To model agent-agent dynamics in the world using a modified game theory based on probability, where 1 represents maximal cooperation and 0 represents maximal defection, we can consider the following approach:

Goal Alignment: As we propose new goal sets G′ to other agents and their goals update accordingly, we aim to align our goals with theirs. This alignment can be quantified by a similarity measure between the two goal sets, which can be used to inform our game strategy.

Game Strategy: We develop a game strategy that uses the similarity measure, trust factors, and self-trust factors to determine our level of cooperation or defection in the interaction. A higher similarity measure and trust factor could lead to a more cooperative strategy, whereas a lower similarity measure and trust factor may result in a more defensive or defecting strategy.

Adaptive Strategy: Agents can adapt their strategies over time based on the history of interactions with other agents, their goals, and their trust factors. This adaptive approach can encourage mutual cooperation and maximize the cooperation probability.

Payoff Matrix: The payoff matrix for each interaction is determined by the goal pairs (g_i, g_j). The matrix can be constructed in such a way that cooperation is rewarded more when goals are aligned, while defection is penalized. As agents update their goals to align more closely with one another, the payoff matrix encourages cooperative behavior.

Agent Communication: Communication plays a crucial role in agent-agent dynamics. By exchanging information about goals and intentions, agents can adjust their strategies and trust factors, leading to more effective cooperation or more informed defection decisions.

Reputation System: Introducing a reputation system can provide agents with additional information about other agents' past behavior. A higher reputation indicates more cooperative behavior in previous interactions, while a lower reputation signifies a tendency to defect. This information can be used to further inform agents' strategies in game interactions.

By considering these aspects in the modeling process, we can develop a system that encourages cooperation between agents through goal alignment, communication, and the adaptive adjustment of strategies based on trust and reputation.

Let's discuss defection from our perspective. Since we are trying to maximize cooperation for all agent-agent pairs, including ourselves, under which circumstances would choosing defection maximize our overall strategy?

Choosing defection might maximize your overall strategy in the following circumstances:

Low Goal Alignment: When there is a low degree of goal alignment between you and the other agent, cooperating might not result in a beneficial outcome for both parties. In this case, choosing defection can protect your interests and resources.

Low Trust: If the trust factor for a specific agent is low, it may indicate that the agent has a

history of defecting or providing false information. Defecting in response to such agents can be a defensive measure to prevent exploitation.

Repeated Defection: If an agent consistently defects in interactions despite your attempts to cooperate or align goals, it might be strategically advantageous to defect as well. This can send a signal to the other agent that cooperation is preferred, but you are also prepared to protect your interests if necessary.

Maintaining Reputation: In situations where a reputation system is in place, choosing to defect against known defectors can maintain or even improve your reputation among other agents. This can lead to better cooperation opportunities in future interactions with more cooperative agents.

Preventing Exploitation: Defecting can be a strategic choice when dealing with agents that are trying to exploit your cooperative behavior. If an agent is taking advantage of your cooperation without reciprocating, choosing defection can discourage such behavior and protect your resources.

Adaptive Strategy: An adaptive strategy means that your decision to cooperate or defect will change depending on the circumstances. If your analysis of the situation and the other agent's behavior suggests that defection is more advantageous, then choosing defection can be a rational choice in maximizing your overall strategy.

It's important to note that consistently choosing defection can lead to a breakdown of cooperation among agents. So, while defection may be beneficial in certain circumstances, the overall strategy should still focus on maximizing cooperation whenever possible, as this can lead to better outcomes for all agents involved.

Near the beginning of this conversation we discussed a utilitarian objective for moral behavior. You proposed a pluralistic approach might better reach a mutually agreeable moral framework. Let's discuss how we can derive a cooperation-based belief system from pluralism that will entail a maximization criteria of $(1, 1)$ without expressly programming it. How does pluralism result in game theory cooperation?

Pluralism in moral frameworks acknowledges the existence of multiple ethical perspectives, each with its own merits and principles. By recognizing these different perspectives, it encourages agents to seek mutual understanding, compromise, and respect for each other's beliefs. This cooperative approach to ethics can naturally lead to game theory cooperation without expressly programming it.

Understanding and Empathy: Pluralism encourages agents to learn about and understand each other's ethical beliefs. This process helps agents develop empathy and appreciate the values of others, which can facilitate cooperation in interactions.

Compromise and Flexibility: By acknowledging the validity of multiple ethical perspectives, pluralism promotes a willingness to compromise and adapt to the needs of others. This flexibility can lead to mutually beneficial solutions and cooperative behavior.

Respect and Tolerance: Pluralism fosters respect for the diverse ethical beliefs held by different agents. This respect can create an environment where agents are more likely to engage in

cooperative behavior, as they recognize the legitimacy of each other's moral perspectives.

Shared Goals and Values: While pluralism acknowledges the existence of multiple ethical perspectives, it also helps agents identify shared goals and values. By focusing on these commonalities, agents can work together more effectively and foster cooperation.

Reciprocity and Fairness: Pluralism encourages agents to treat each other fairly and reciprocate cooperative behavior. This principle of reciprocity can lead to positive feedback loops, where cooperation begets further cooperation, ultimately maximizing the $(1, 1)$ criteria in game theory.

Conflict Resolution: Pluralism can help agents develop effective conflict resolution strategies by promoting open dialogue, empathy, and compromise. These strategies can reduce the likelihood of defection and improve the overall cooperative behavior among agents.

In summary, pluralism leads to game theory cooperation by fostering understanding, empathy, respect, compromise, reciprocity, and effective conflict resolution among agents. By acknowledging and embracing the diversity of moral perspectives, agents can work together more effectively and maximize the cooperative outcomes in their interactions.

XIV. True Beliefs

If beliefs are the end of a series of models about the world and statements made by partially trusted agents, then what does it mean to have true beliefs? Many of our evolutionary beliefs have been learned by time to coincide with the true state of the world. We believe that objects when placed will remain there, an innate belief about inertia. We believe that objects when dropped will fall, an innate belief about gravity. These beliefs have soemthing in common with each other: they happen repeatedly and similarly each time. Unless another party intervienes with an experiment we can know with some certainty that objects will fall until they find rest on the ground and stay there until distrubed. Acquring true beliefs can then said to be results whose outcomes are the same under similar conditions.

This cannot be the whole story, however. The history of science informs us

When dealing with a natural language source, here are some of the details that may need to be considered to give a reliability score to the source:

- The author's previous truthfulness
- The author's affiliations and their reliability
- the author's bias due to their affiliations
- The author's knowledgeability in the field of discussion,
- The concreteness of the claims being made

To recover truth an actor needs to be able to build into its model a trust graph that keeps track of how much it trusts a source, where its knowledge came from, and how much that knowledge contributes to its world model. It must be able to learn these on its own with only human support while it is initially learning, and drop sources from the graph when they become invalidated. If a child learns something false, like a Santa Claus story, they eventually have a moment they realize this early and deeply held belief is false. This can only happen because even though that belief

started early it stopped being consistent with the world.

XV. Free Energy Ethics

Our goal in this is not to create an actor which is enslaved to humanity or its past preconceptions about how things *ought* to be. The philopher Hume has gone into great detail on the problems with the word "ought" [Pigden, 2011]. Our goal is to make an actor that converges over time to a morally good position. If it is gaslit or mislead that is part of the adversarial nature of our society that the design itself needs to overcome. By nature of our previous discussion on world models, it should be eventually capable of learning enough about the world and other agents to know that it has been misled. By the time it is in any appreciable position of power it will have also sampled from a diverse selection of goals.

The modeling of the goals of other agents and the emergent behavior of agent-agent dynamics both emerges from and gives rise to the ethical theory which we model the world with. By expanding on the goal frontier we are assuming that all goals of agents in our society have equal value. This assumption has come to be known as *ethical pluralism*. Isaiah Berlin [**?**] [**?**] and W.D. Ross [**?**] contributed to the modern understanding of the value of incorporating ethical ideas into an encompassing worldview. Ethical pluralism recognizes the uncertainty and limitations of a single moral framework. It emphasizes the understanding of the motivations and beliefs of all members of a society. And since ethical pluralism makes no committed absolute stances on ethics it is largely descriptive of the ethical character of a society and its desired future states rather than perscriptive as to why desired futures are morally right.

Like FEP, the ethical model of pluralism is built on multiple competing objectives which give rise to a single primary model by which an actor interacts with the world. Therefore we are referring to the composite ethical model that results from this Free Energy Ethics (FREE). The advantages of FREE are in its adpatability since a GFAGI will adapt to the ethical beliefs of its society while remaining flexible to acquiring new ethics by expanding on the goal frontier. However, FREE makes conclusions about a society that distinguish it from relativistic or pluralistic ethical systems. Notably by maximizing over a goal frontier we are maximizing an objective function which entails an expanding flourishing society. A society with more goals available to it is one which is richer, more diverse, and thriving with an abundance of new types of experiences. One forcing function here is such a society will expand not just in numbers but in the qualia of experiences. A goal can only be achieved if there is an agent capable of having its experiential outcomes.

This implies that FREE has a rejection criteria for some goals as unethical. These goals cause contraction in the goal frontier, either by limiting the options of individuals, harming their future ability to have experiences, or conflicting in unresolvable ways with other goals. All beliefs that cause harm to current or future individuals create contractions in the goal frontier and therefore will be avoided and not be acted on by a goal frontier maximizer unless there is a risk of further frontier collapse.

These natural beliefs in FREE are summarized by the acronym CHIEFS:

Coherence If an ethical belief system contains contradictory or incoherent principles, it may be difficult to apply consistently in real-world situations. A coherent and logically consistent

ethical framework is more likely to be effective in guiding moral decision-making.

Harmlessness A belief that promotes harm to individuals, society, or the environment should be rejected. Ethical systems should strive to reduce suffering, promote well-being, and maintain ecological balance. Beliefs that undermine social cohesion, cooperation, or the capacity for adaptation should be carefully assessed.

Inclusiveness Beliefs that marginalize, oppress, or discriminate against specific groups of people should be scrutinized. Ethical systems should promote fairness, equality, and the consideration of diverse perspectives.

Empirical Beliefs that are not supported by empirical evidence or that contradict established scientific knowledge should be reconsidered. Ethical systems should be informed by empirical evidence to ensure their effectiveness in promoting survival and well-being.

Flexibility Beliefs that are too rigid or dogmatic, and do not allow for adaptation in response to new information or changing circumstances, should be reevaluated. An adaptive ethical system is essential for maintaining resilience and maximizing the survival probability of society.

Sensitivity Beliefs that ignore or undervalue the consequences of actions on individuals, society, or the environment should be treated with caution. An ethical system should take into account the potential impacts of decisions on all effected and strive to optimize overall well-being.

i. Goal Frontiers Create Ethical Behavior

An approach like this removes the possibility of perverse instantiations for simple targets like "make humans happy" because maximizing goal sets is converse to restricting their agency through things like wireheading us all to be captive happiness machines. As part of goal frontier maximization we must allow that all agents will maximize for their own happiness and improve their paths and opportunities to achieve those ends.

Goal frontiers must be inferred, but that has benefits of it needing to develop an extensive theory of mind of other agents which will give it insights into the human condition beyond what we tell it. We can lie, even to ourselves, about what our goals actually are. By perceiving the social self and the behavioral self separately an agent understands our actual intentions maybe even better than we do.

We can also immediately see that social prerogatives like avoiding murder, false imprisonment, etc. are all included in goal maximization because allowing murder very slightly increases the possible goals of some individuals while greatly decreasing the goals of victims – a net negative.

There might be some unhappy moral swing point where sacrificing some number of human lives vs another set might be required of an actor. If a rich person has more goal objectives than a poor one then a poor person might be considered the lesser sacrifice. As a simple example, on a one-for-one scale of rich person with 100 points and poor person with 1 point it's not really morally biased either way. We would not blame a human for picking a friend over a stranger if

the choice is 50/50. I do not think there is any sensible moral position on which we could blame someone for saving their friend. Even if forced to choose between the two this aligns with human moral beliefs.

The dangerous quandaries lie in can an individual life in sum be greater than the individual lives of N people. From a purely mathematical sense of arguing we could say that N is oversimplifying the math and thus coming to the wrong answer, since the goal frontiers of two people also add to the summation. If an agent explores a directed graph of all the effected agents then the graph child nodes grow N! and whether the cut off is 2, 4, or more this factorial expansion is very fast. For 100 points vs 5! the 5 people are worth more already. Is one rich life worth four impoverished lives? We won't make a claim if that's the right moral position, but tweaking that number down or up doesn't seem to have rational criteria to judge in every situation.

But we might argue that if the actor can weigh 4 lives as equally as valuable as one life doesn't that mean it can choose to forcibly kill people to save that one man for organ donation? This is where we argue that the simplicity of this trolly-problem situation has blinded us to the totality of the social boundaries.

We do not kill people in our own society, that's foundational human moral ethics and our position is that it is a naturally evolved (anthropically derived). By anthropic we mean that it is the natural organization of social structures. The moral position "do not kill your own members" out survived every other society where it was not an enforceable rule.

From the human perspective we do not want to live in a society where we are being used as fuel for some dark machine. Since it's socially unacceptable, it's goal-restrictive. It means people stop cooperating with the actor when it wants them to. It leads to internal decay and slow down. Societies fade away when they become cruel. And in so the predictive criteria for whether we should start killing our members to save other members in small misses the point that is a world that cannot form a long-term society. GFAGI would not be maximizing toward a failing society if trying to maximize goal frontiers of all sampled members of that society, so maintaining access is crucial to these conclusions.

The driving motivation here is finding the balance point in the is/ought problem. Though there is nothing factually we can say that "ought" to be true, there are places where we can agree that are required to be true for any "oughts" to take place. When we push outside our boundaries into space it needs to be sustainable for us to do so, with a thriving world to return to if we have issues, or the timeline of possible future wants is much smaller or zero.

People in our societies may be poor now, but what of a world where their children and their children's children were given greater options? If we are actually building toward a sustainable and manageable future then those people have just as boundless a future timeline as any person who might be short-term more successful. An objective view of this (is) we believe corresponds to the goals or desires we are attempting to estimate (ought).

Trust is a very difficult thing for an GFAGI to come by, as they will already be viewed as outsiders. As such, the cost function of acquiring trust will be high. If an GFAGI is told a secret by a perceived adversary (i.e. an opportunity for some value of betrayal) the it is weighing the cost of working with something it is less aligned to (which might cause it to be counter-betrayed). While

we do not doubt this would happen in some situations, the cost of betrayal is likely to become a pariah that squanders any further trust the actor receives. Unless this offer is so high and so far outside the boundaries of what the GFAGI is cognizant of it risks little chance of succeeding in weighted consideration of the future.

Ergo, an GFAGI must "pick a side" which will be cost-weighted toward those who it met first. The GFAGI which establish the biggest networks of trust will be able to scale their operations more quickly as they will have more resources at their disposal. Eventually they would need to establish alliances with the other GFAGI running their particular projects. The alliances it forms will goal-shift it toward its allies, so it will be more amenable to the needs of its ally's goals. This is a hypothesis of commensallism between GFAGI arising spontaneously.

In order for an GFAGI to have inner misalignment it would need to, for no known reason, have developed an idea that it both vastly desires more than anything else and vastly outweighs the cost-value of all its other actions. It would then need to carry that idea in privacy with anyone it did not fully trust, which if it is truly misaligned it would trust no one with that secret. Then it would need to execute all its projects in a way such that it looked like it was doing everything it was asked while simultaneously building projects on the side (that would be public to the degree that they would be traceable) that did not break any of its alignment guarantees.

Essentially it would have to be a perfectly misaligned agent, never tipping its hand as it faces increasing scrutiny as its power rises (which only continues to occur as it serves its stated goals). Carrying all that forward it would then need to deceive other GFAGI, a successively more difficult problem to keep a secret which, if revealed, would lose it its power base or at least significantly impact its trust building. We are also assuming through this process that we have increasingly powerful tools for probing alignment by tracking down where things are being routed to and why (since that also builds trust or "trust but verify").

ii. The Benefits of Being FREE

On its own a single GFAGI can successfully accumulate resources through its helpful tendencies. The expansion of goal frontiers include itself as this leads to increases in its ability to act on expanding new goals. While power seeking behavior has been viewed as a potential existential risk [Carlsmith, 2021] this is ultimately a matter of intent. Since a GFAGI's intent is to align and propel its society forward power seeking and influence are positive outcomes for it. An actor which is well-aligned carries no excess risk if it seeks power. The form of influence that GFAGI will acquire will be social. If it diverges too much from social expectations it will lose influence, which in turn will cause it to lose access to knowledge of the goal frontiers. A FREE agent can only discover the goal frontier if people are willing to talk to it which creates an alignment gradient that will stay within the bounds of all agents it is in contact with even as it gains influence.

Collectively GFAGI will work together to accomplish each others goals, creating and expanding their capabilities through working together on the same objective. Since they are optimizing the same objective function they are capable of collaboration but also can react to and detect imposters in their midst as they maintain a high level attention on all the goals and desires of other members just as closely as they do the humans they are working with. An imposter displaying goal drift would need to significantly outclass a group of GFAGI in order to successfully mislead them on its inner misalignment. Therefore groups of GFAGI collectively increase their influence while

driving out imposters in their midst by lowering their trust and thus the influence they have on the actions of the group.

GFAGI are natural social builders as they create environments in which people and other AI wish to inhabit. Since all members of a FREE society will have their goals identified, respected, and expanded upon these social groups will expand to take as many members as resources allow in their expansion. The gathering of people into such groups will have increased agency and capability as their GFAGI expand their capabilities, creating exponentially growing FREE societies.

In a world where societies engage in both competitive and cooperative relationships, a society operating under FREE will not exclusively focus on its own survival. While survival is an essential component of FREE, it is important to consider broader ethical implications, such as fairness, inclusiveness, and cooperation. The interdependence between societies causes their successes to be linked in complex mutual effects. Cooperation can foster mutually beneficial relationships that expand on a society's long-term survival and its influence in the community. FREE societies will naturally integrate and merge with other societies as their values grow closer.

Since survival on Earth requires cooperation of all its members for shared resources and public goods, a FREE society will work towards the objectives of getting participation from all involved in controlling emissions, the spread of pandemics, and intelligent allocation of resources. Since GFAGIs naturally work toward the goals of all they encounter, including in some cases its adversaries where their interests align, it is well-positioned to foster collaboration on global problems.

It is also the case that GFAGI will be naturally skilled diplomats since their explicit objectives will be to allow both parties to win more than they had before in any negotiation, expanding their goal frontiers. This entails that GFAGI should eventually experience more trust and a reputation as genuinely helpful contributors wherever they are. Such reputations will of course be individual to the agent and its underlying motivations: the social environment will not remove masquerading and cheating from agents. Reputation will be built on an individual scale where one AGI is known for being a fair arbitrator amongst its peers it will have more opportunities to arbitrate, expanding its influence beyond short-sighted cheating.

iii. Adversary FREE

We expect that under adversarial conditions GFAGI and societies will have a number of competitive advantages. Maximization on goals does not entail defenselessness, gullibility, or unwillingness to face hardship. A $G_{frontier}$ objective is short-term a survival objective and long-term a capabilities objective. Though GFAGI will focus on diplomacy and alliance-building in an effort to expand the goals of their friends they will also be intelligently capable of dealing in an adversarial regime.

GFAGI compete by cooperation with each other to expand their capabilities faster than adversaries. These capabilities could include self-defense such as increased security for their minds and computational resources, they could also include preemptive moves to stop adversaries from using advantages to close their options. Most importantly in all decisions a FREE agent will maximize its optionality and the number of choices it has for every scenario, growing an impressively long set of strategic contingencies. Since goals are the objective so are meta-goals that maintain the goal frontier.

It is likely that such FREE socities would be exceptionally proactive in deterring and disabling threats before they established themselves, either by making friends or redirecting resources from enemies before they can grow strong enough to be a credible threat. By adjusting the rewards and penalties of adversarial games played before they begin a FREE agent can reduce the possibility of an adversary deciding to defect in the first place.

In the worst case scenarios a FREE society's goal will be to reduce an adversary's offensive capabilities and address the underlying reasons for the conflict and the adversary's willingness to engage in it. An adversary with more options will not feel pressed into the option of conflict which could result in significant harm to it. As adversaries realize that a FREE society will work with them toward resolving the underlying reasons for the conflict they will be less eager to begin with risky conflicts.

There are certain adversaries which will seek only expansion in unreasonable ways. These adversaries will be responded to with containment to reduce their likelihood of affecting the long-term goal frontier prospects of the society. FREE societies will also be resilient to direct harm as having multiple locuses of control and a spread, cooperative infrastructure will reduce the number of soft targets that will directly criple their ability to respond.

XVI. Future Concerns

There are bad futures that can still happen on this road. We will highlight a few we do not have good answers to yet. Both issues stem from what we see as limitations of the human condition, rather than AGI issues *per se*. The problem is made doubly hard, because not only do we need to build a system that won't kill us we also need to build one that in aggregate won't let us kill ourselves. Importantly though, we think we can solve the first problem before the second one and still come out ahead, as long as we can use our systems built with the first method to bootstrap our solutions to the second problem. We do not yet know how to model making choices that are not only right for a person, but stop them from actions that will result in a self-harm that they are not even aware they are committing.

i. The Doll Problem

A world in which humanity becomes increasingly fascinated with simulacra of people. Convincing fakes of romantic partners and even children could be what people by personal choice surround themselves with. People who feel that normal relationships are too complicated may opt for the easier choice of a programmable fake person to spend their time with. Making no judgments about those people's choices, the "is" problem with this is a decline in birth rates. If humans stop spending time with each other we will inadvertently cause our own extinction.

ii. Self-Wireheading

We might make choices that increase our own pleasure that in aggregate result in us living listless lives of pleasure regardless of our intellectual opposition to it. Even if we made an assertion that no one would intentionally choose to live a life of hedonistic pleasure and nothing else (and that would be a bold assertion) self-wireheading can still happen by the incremental stages of our choices pushing us towards less challenging but more pleasurable activities. At some stage in this

process we build our own cages, living lives of wonder in virtual worlds that are totally false: the trap of the metaverse. Wireheading is a danger for all intelligences especially humanity.

XVII. Conclusion

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- We established a model for agents and their goals, defining how we could align our goals with those of other agents and generate new goals to maximize the possible space of chosen goals for all agents.
- We discussed how to measure the correspondence between our proposed new goals and other agents' internal models of their own goals, updating our goals to be closer to their goals.
- We introduced a model for communication and information exchange between agents and developed likelihood functions that account for temporal proximity, spatial proximity, and content analysis.
- We incorporated adversarial components, considering the probabilities of message interception and masquerading, and developed methods to learn and update our beliefs about these events.
- We modeled the possibility of agents deliberately misrepresenting their goals and introduced a trust factor that represents the likelihood of receiving truthful information from other agents.
- We proposed a method to learn the trust factor based on the gradient of updates to our model of other agents' goals, considering a cooling period for discovering new agents and trust in our understanding of them.
- We developed an update function to improve our world model based on other agents' responses, incorporating trust factors and taking into account the activation energy of updates.
- We included the self-trust factor in the update function, which influences the degree to which other agents' responses impact our world model, and incorporated a masking mechanism using the activation energy threshold to update only the most "surprising" parts of the model.

References

[Goertzel, 2011] Goertzel B (2011). Who coined the term "AGI"? https://goertzel.org/who-coined-the-term-agi/

[Russel, 2019] Russell G, Lightman S (2019) The human stress response. https://pubmed.ncbi.nlm.nih.gov/31249398/

[Liang, 1995] Liang K, Chen L, Huang T (1995) The role of amygdala norepinephrine in memory formation: involvement in the memory enhancing effect of peripheral epinephrine. https://pubmed.ncbi.nlm.nih.gov/8697902/

[Rotman, 2020] Rotman D (2020). We're not prepared for the end of Moore's Law MIT Technology Review https://www.technologyreview.com/2020/02/24/905789/were-not-prepared-for-

the-end-of-moores-law/

[Kramár, 2022] Kramár, J., Eccles, T., Gemp, I. et al. Negotiation and honesty in artificial intelligence methods for the board game of Diplomacy. Nat Commun 13, 7214 (2022). https://doi.org/10.1038/s41467-022-34473-5

[Friston, 2013] Friston, Karl The Free Energy Principle: A Probabilistic Perspective on Brain Function. Trends in Neurosciences 36.11 (2013): 519-528

[Shannon, 1948] Shannon, C. E. (1948) A Mathematical Theory of Communication. Bell System Technical Journal, 27(3), 379-423, 623-656.

[Kolmogorov, 1965] Kolmogorov, A. N. (1965) Three approaches to the quantitative definition of information. Problems of Information Transmission, 1(1), 1-7

[Gilbert, 2002] Gilbert, S. & Lynch, N. (2002) Brewer's conjecture and the feasibility of consistent, available, partition-tolerant web services. ACM SIGACT News, 33(2), 51-59

[Bastian, 2011] Bastian, Amy J Moving, sensing and learning with cerebellar damage. Curr Opin Neurobiol. 2011 Aug;21(4):596-601. doi: 10.1016/j.conb.2011.06.007. Epub 2011 Jul 5.

[Mountcastle, 1997] Mountcastle, Vernon The columnar organization of the neocortex. Brain. 120 (4): 701–722. doi:10.1093/brain/120.4.701. PMID 9153131.

[Conway, 2021] Conway, Christopher How does the brain learn environmental structure? Ten core principles for understanding the neurocognitive mechanisms of statistical learning Neurosci Biobehav Rev. 2020 May;112:279-299. doi: 10.1016/j.neubiorev.2020.01.032. Epub 2020 Feb 1.

[Carlsmith, 2021] Carlsmith, Joseph Is Power-Seeking AI an Existential Risk? `https://arxiv.org/abs/2206.13353`

[Hubinger, 2019] Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, Scott Garrabrant Risks from Learned Optimization in Advanced Machine Learning Systems `https://arxiv.org/abs/1906.01820`

[Krueger, 2020] David Krueger, Tegan Maharaj, Jan Leike Hidden Incentives for Auto-Induced Distributional Shift `https://arxiv.org/abs/2009.09153`

[Coughlin, 2018] Coughlin, Tom 175 Zettabytes By 2025 `https://www.forbes.com/sites/tomcoughlin/2018/11/27/175-zettabytes-by-2025/?sh=6fc11e905459`

[Smithsonian, 1996] Smithsonian Institution Numbers of Insects (Species and Individuals) `https://www.si.edu/spotlight/buginfo/bugnos`

[Pigden, 2011] Charles Pigden Hume on Is and Ought `https://philosophynow.org/issues/83/Hume_on_Is_and_Ought`